**Al-Aqsa University**

**College of Engineering**

**Computer Engineering & Control Systems Dept**

**جامعة الأقصى**

**كلية الهندسة**

**قسم هندسة الحاسوب وأنظمة التحكم**

# smartphone programming (ENGS4301)

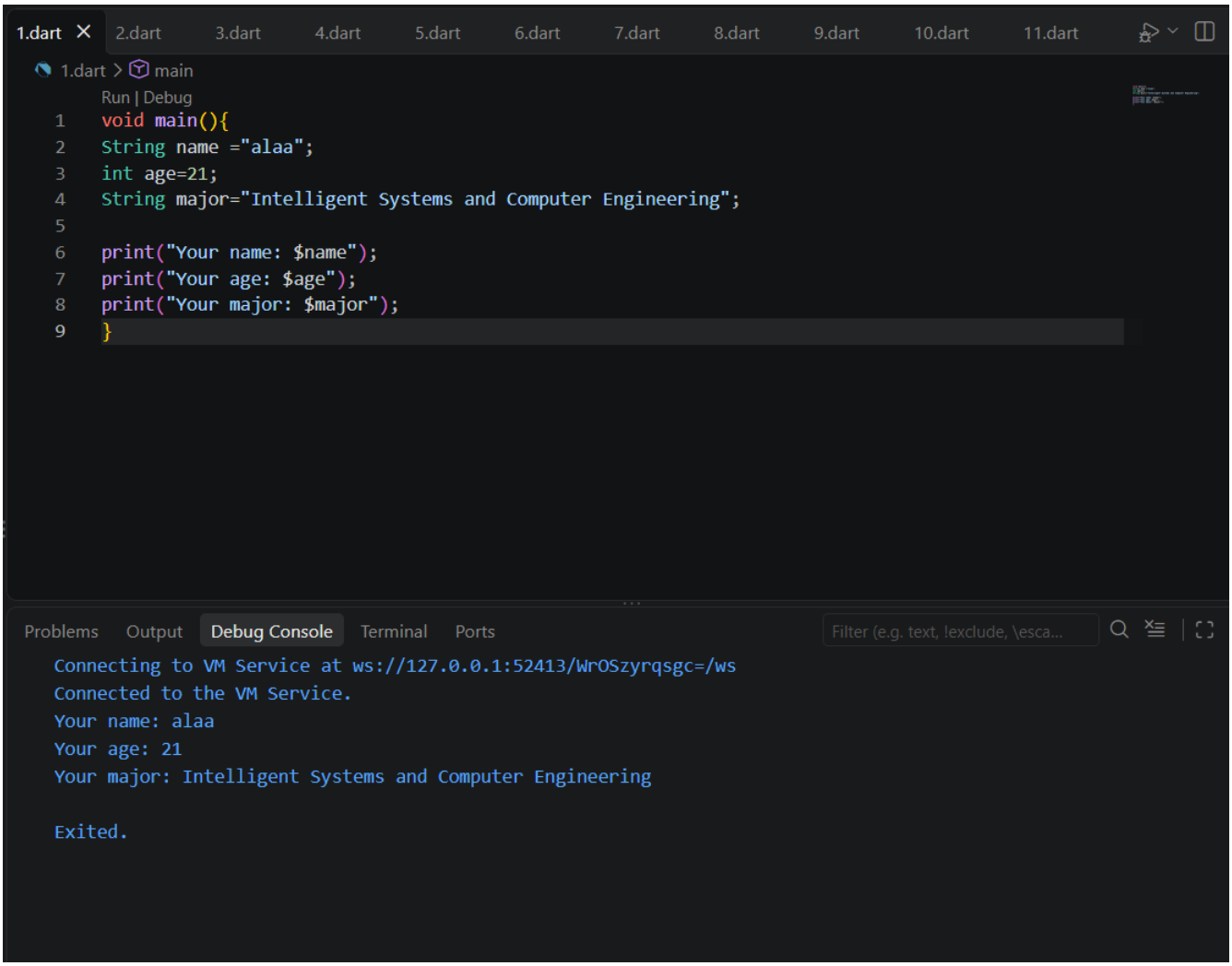
## Lab1

**Student name**: Alaa Al-Ramlawi

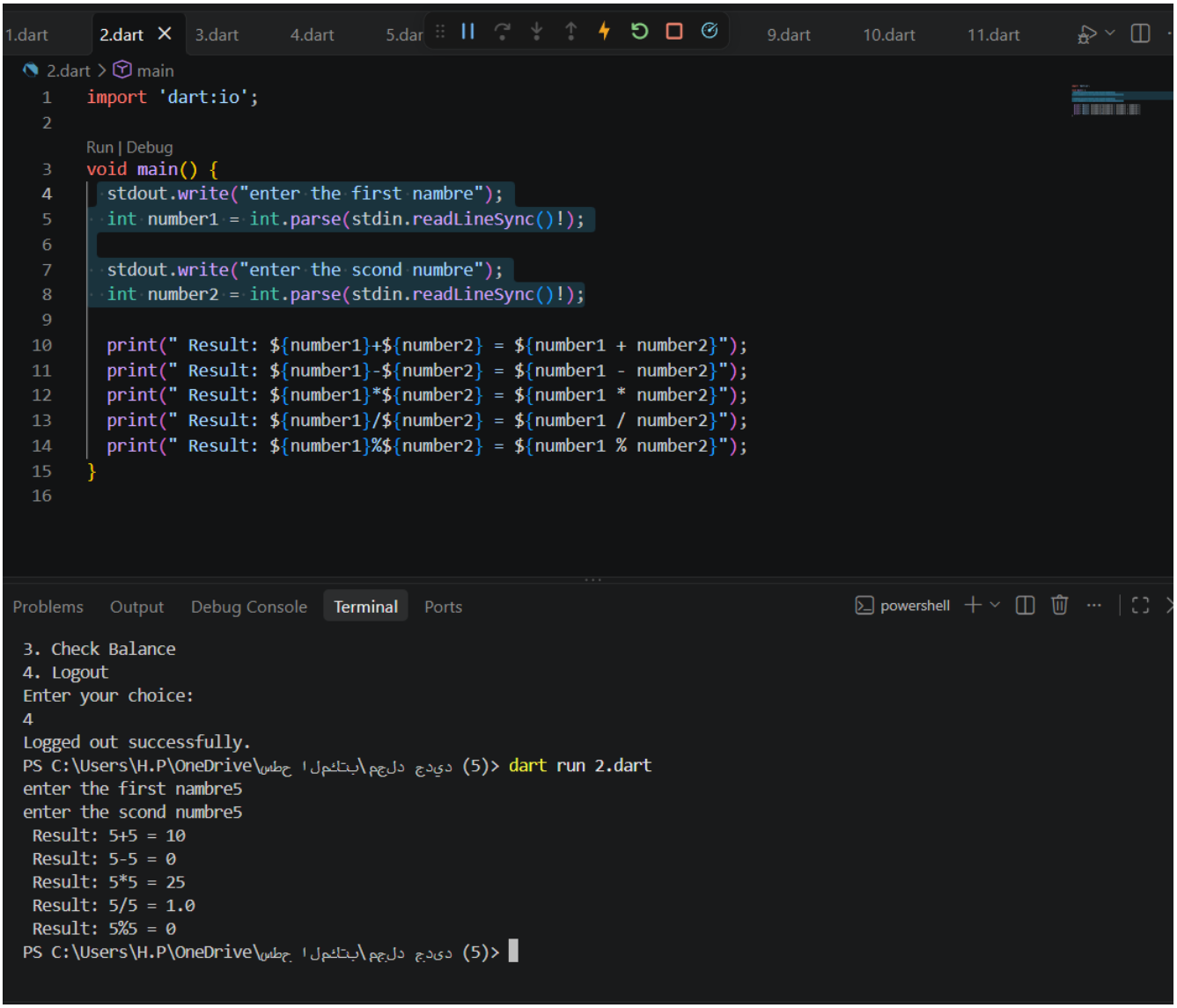
**Presented to**: Eng. Mohammad Iyad Al-Lahham

**Semester**: Second Semester 2026/2027

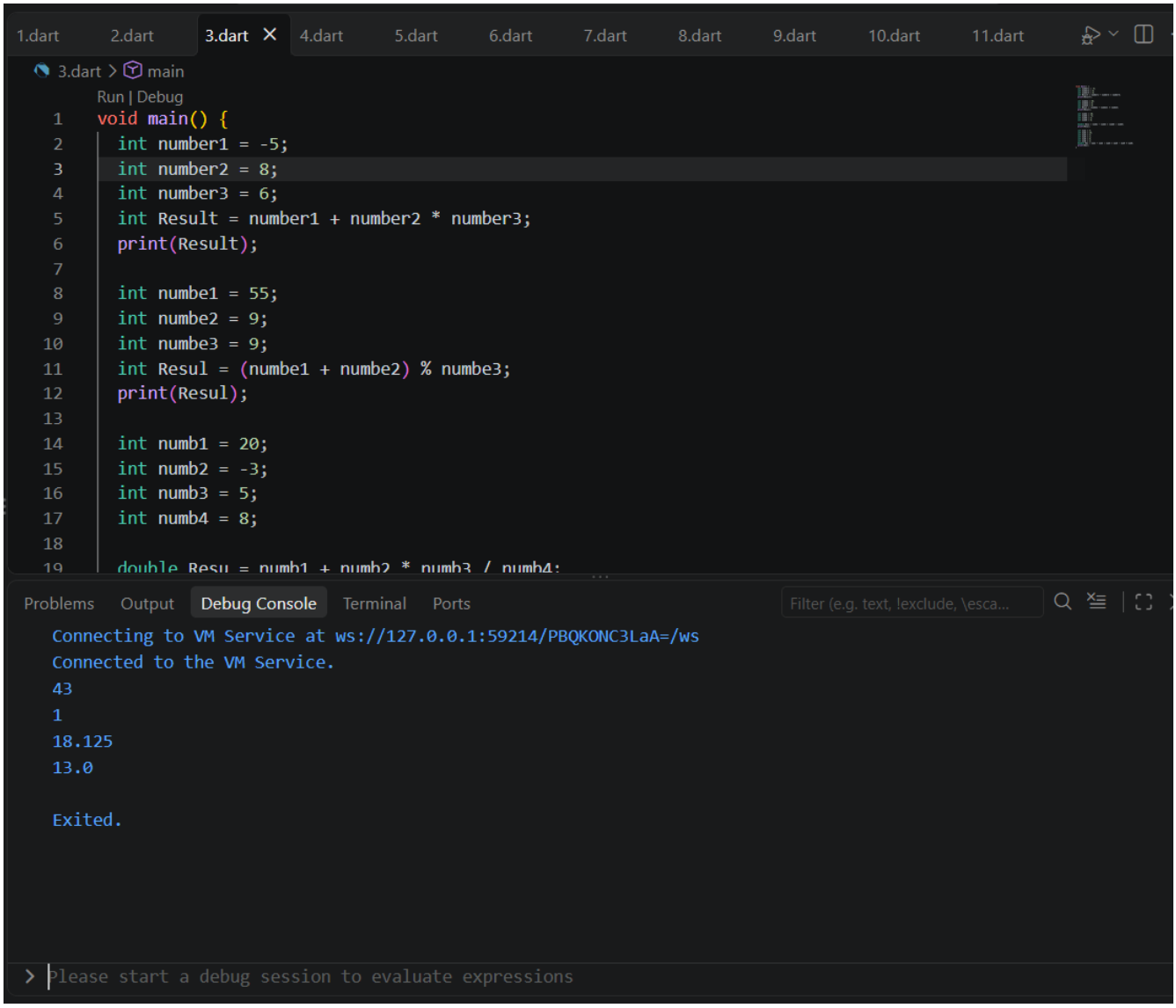
# Question:1


1.dart
2.dart
3.dart
4.dart
5.dart
6.dart
7.dart
8.dart
9.dart
10.dart
11.dart
1.dart > main
Run | Debug
1 void main(){
2 String name ="alaa";
3 int age=21;
4 String major="Intelligent Systems and Computer Engineering";
5
6 print("Your name: $name");
7 print("Your age: $age");
8 print("Your major: $major");
9 }
Problems
Output
Debug Console
Terminal
Ports
Filter (e.g. text, !exclude, \esca...
Connecting to VM Service at ws://127.0.0.1:52413/WrOSzyrqsgc=/ws
Connected to the VM Service.
Your name: alaa
Your age: 21
Your major: Intelligent Systems and Computer Engineering
Exited.

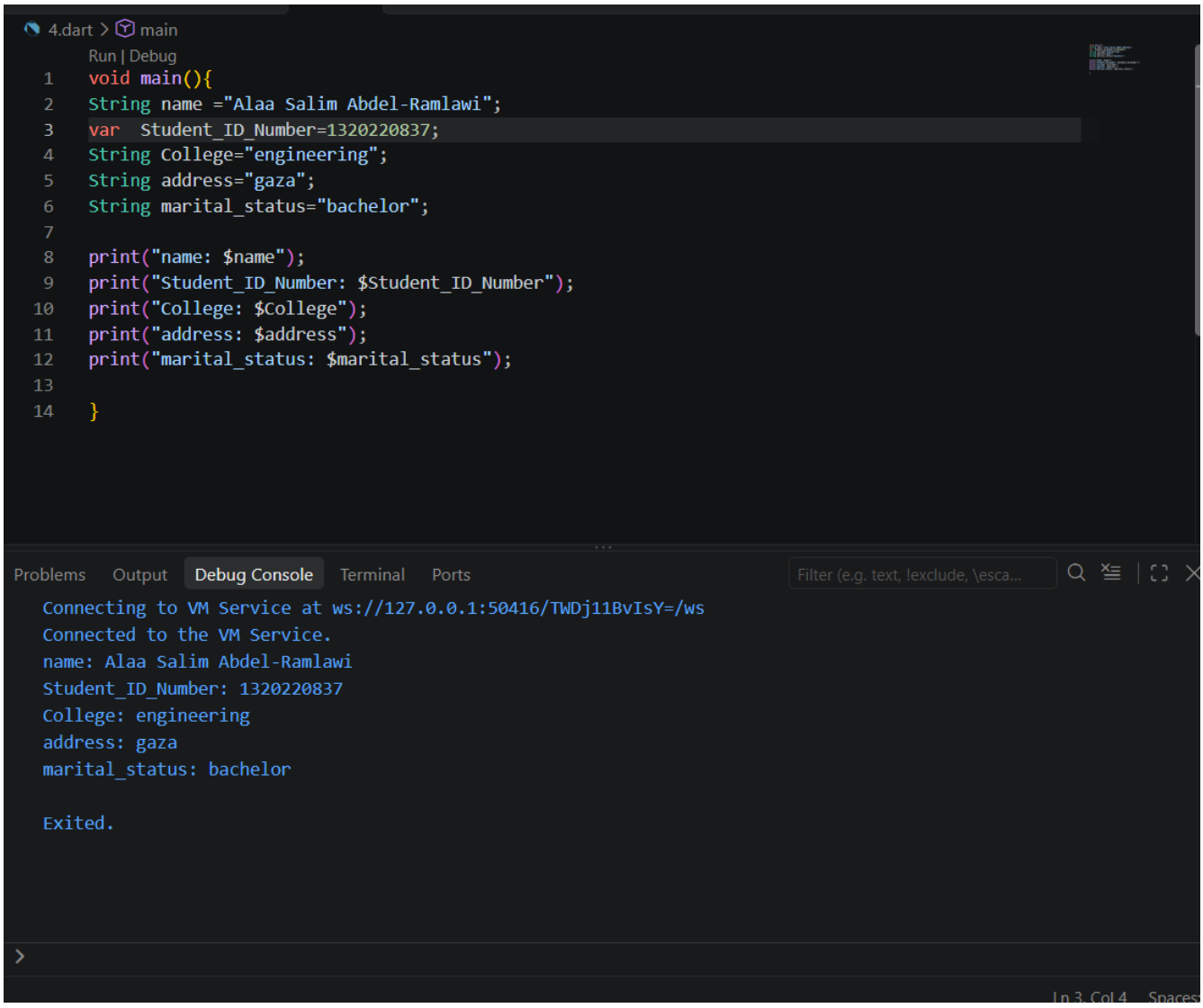
## Question:2


1.dart
2.dart
3.dart
4.dart
5.dar
9.dart
10.dart
11.dart
2.dart > main
1 import 'dart:io';
2
Run | Debug
3 void main() {
4 stdout.write("enter the first nambre");
5 int number1 = int.parse(stdin.readLineSync()!);
6
7 stdout.write("enter the scond numbre");
8 int number2 = int.parse(stdin.readLineSync()!);
9
10 print(" Result: ${number1}+${number2} = ${number1 + number2}");
11 print(" Result: ${number1}-${number2} = ${number1 - number2}");
12 print(" Result: ${number1}*${number2} = ${number1 * number2}");
13 print(" Result: ${number1}/${number2} = ${number1 / number2}");
14 print(" Result: ${number1}%${number2} = ${number1 % number2}");
15 }
16
Problems
Output
Debug Console
Terminal
Ports
powershell
3. Check Balance
4. Logout
Enter your choice:
4
Logged out successfully.
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 2.dart
enter the first nambre5
enter the scond numbre5
Result: 5+5 = 10
Result: 5-5 = 0
Result: 5*5 = 25
Result: 5/5 = 1.0
Result: 5%5 = 0
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>

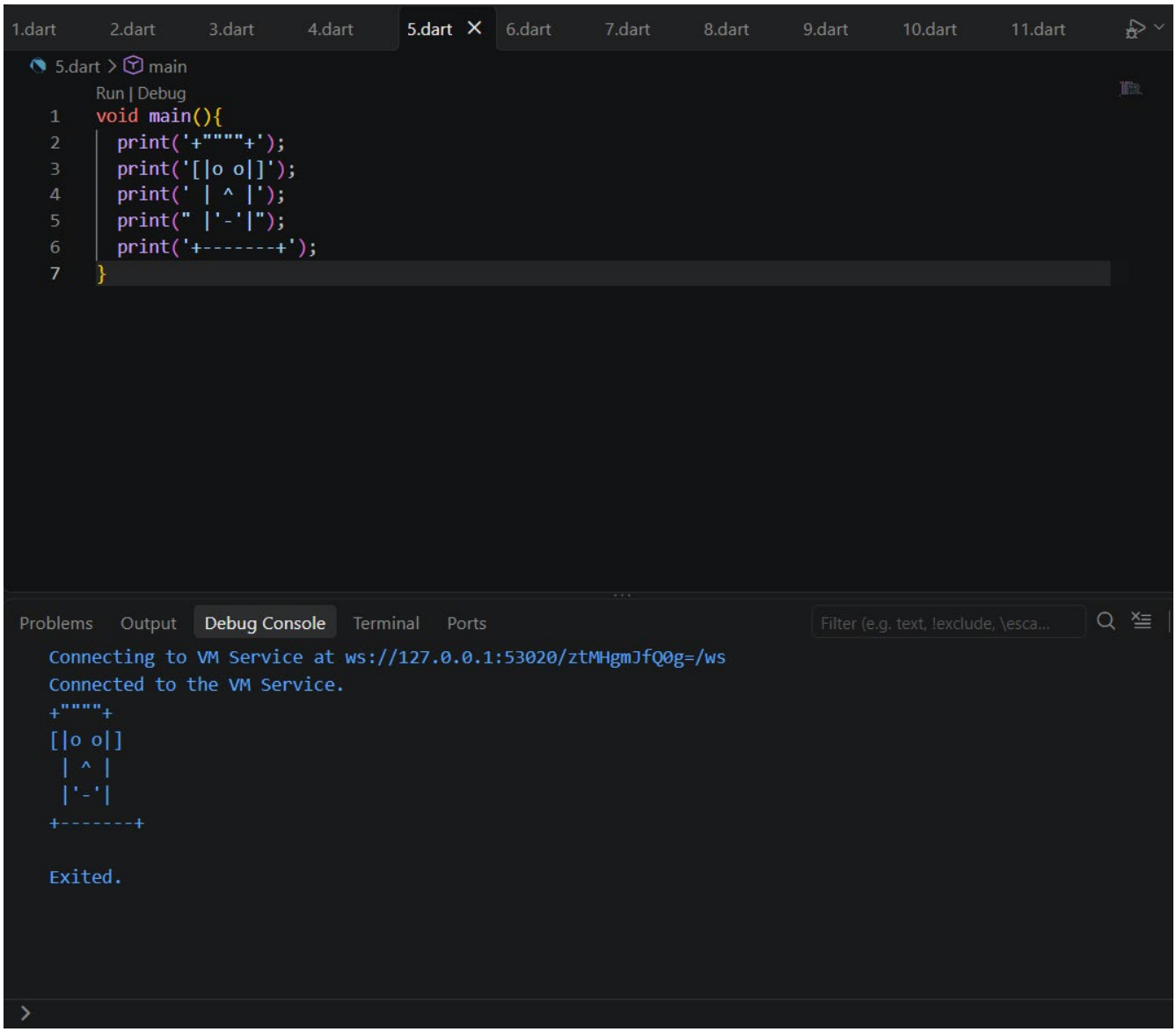
## Question:3


1.dart
2.dart
3.dart
4.dart
5.dart
6.dart
7.dart
8.dart
9.dart
10.dart
11.dart
3.dart > main
Run | Debug
void main() {
int number1 = -5;
int number2 = 8;
int number3 = 6;
int Result = number1 + number2 * number3;
print(Result);
int numbe1 = 55;
int numbe2 = 9;
int numbe3 = 9;
int Resul = (numbe1 + numbe2) % numbe3;
print(Resul);
int numb1 = 20;
int numb2 = -3;
int numb3 = 5;
int numb4 = 8;
Problems
Output
Debug Console
Terminal
Ports
Filter (e.g. text, !exclude, \esca...
Connecting to VM Service at ws://127.0.0.1:59214/PBQKONC3LaA=/ws
Connected to the VM Service.
43
1
18.125
13.0
Exited.
Please start a debug session to evaluate expressions

## Question:4


4.dart > main
Run | Debug
1 void main(){
2 String name ="Alaa Salim Abdel-Ramlawi";
3 var Student_ID_Number=1320220837;
4 String College="engineering";
5 String address="gaza";
6 String marital_status="bachelor";
7
8 print("name: $name");
9 print("Student_ID_Number: $Student_ID_Number");
10 print("College: $College");
11 print("address: $address");
12 print("marital_status: $marital_status");
13
14 }
Problems
Output
Debug Console
Terminal
Ports
Filter (e.g. text, !exclude, \esca...
Connecting to VM Service at ws://127.0.0.1:50416/TWDj11BvIsY=/ws
Connected to the VM Service.
name: Alaa Salim Abdel-Ramlawi
Student_ID_Number: 1320220837
College: engineering
address: gaza
marital_status: bachelor
Exited.
Ln 3, Col 4
Spaces:

## Question:5


1.dart
2.dart
3.dart
4.dart
5.dart
6.dart
7.dart
8.dart
9.dart
10.dart
11.dart
5.dart > main
Run | Debug
1 void main(){
2 print('+"""+');
3 print('[|o o|]');
4 print(' | ^ |');
5 print(" |'-'|");
6 print('+-------+');
7 }
Problems
Output
Debug Console
Terminal
Ports
Filter (e.g. text, !exclude, \esca...
Connecting to VM Service at ws://127.0.0.1:53020/ztMHgmJfQ0g=/ws
Connected to the VM Service.
+"""+
[|o o|]
| ^ |
|'-'|
+-------+
Exited.

## Question:6

```dart
import 'dart:io';
Run | Debug
void main(){
stdout.write("enter the length");
int length =int.parse(stdin.readLineSync()!);

stdout.write("enter the width");
int width=int.parse(stdin.readLineSync()!);

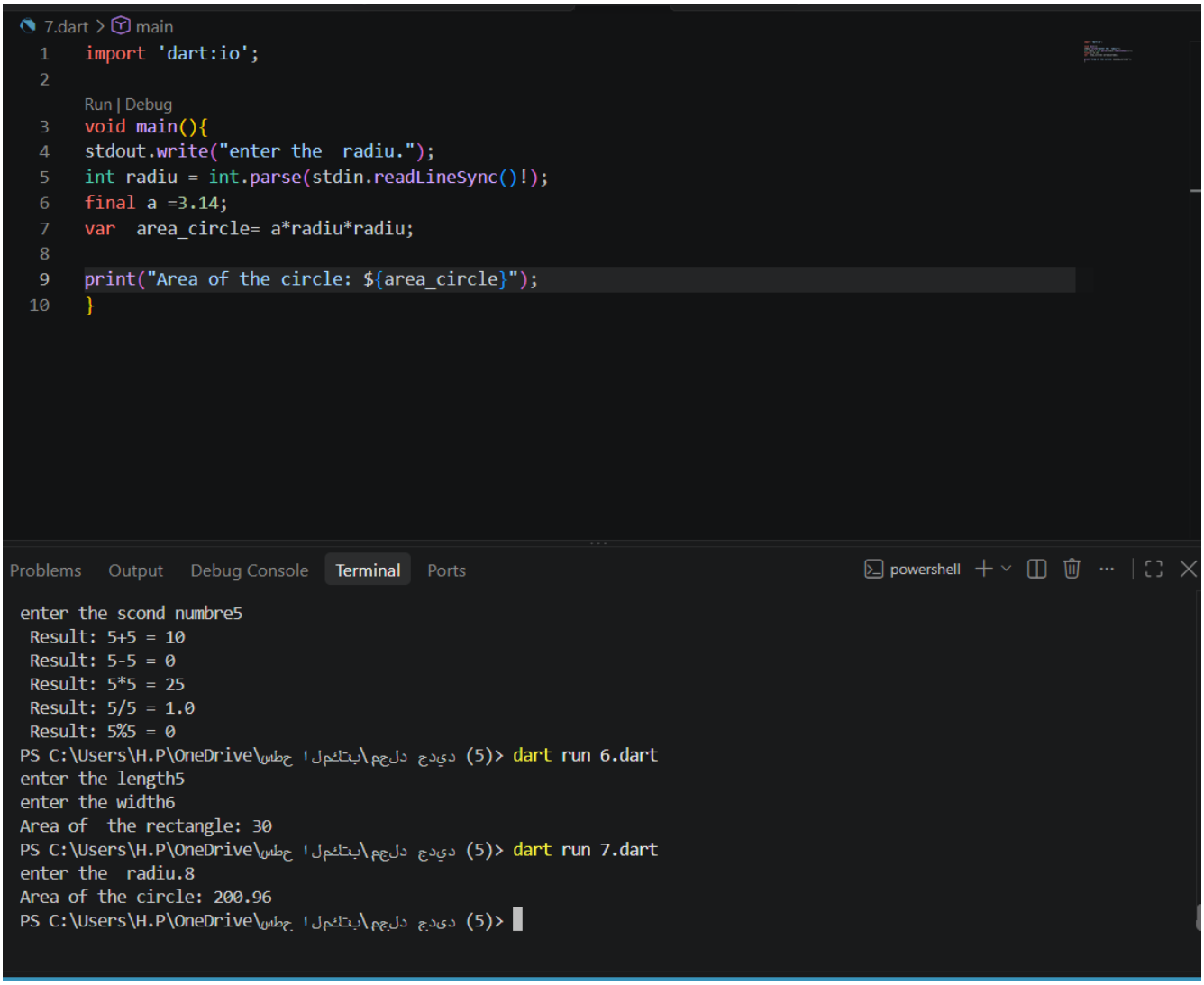
  int Area_rectangle=length*width;

print("Area of  the rectangle: ${Area_rectangle} ");
}
```

```
Problems  Output  Debug Console  Terminal  Ports        powershell

Logged out successfully.
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 2.dart
enter the first nambre5
enter the scond numbre5
 Result: 5+5 = 10
 Result: 5-5 = 0
 Result: 5*5 = 25
 Result: 5/5 = 1.0
 Result: 5%5 = 0
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 6.dart
enter the length5
enter the width6
Area of  the rectangle: 30
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

## Question:7



```dart
import 'dart:io';

Run | Debug
void main(){
stdout.write("enter the  radiu.");
int radiu = int.parse(stdin.readLineSync()!);
final a =3.14;
var  area_circle= a*radiu*radiu;

print("Area of the circle: ${area_circle}");
}
```

```
enter the scond numbre5
 Result: 5+5 = 10
 Result: 5-5 = 0
 Result: 5*5 = 25
 Result: 5/5 = 1.0
 Result: 5%5 = 0
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 6.dart
enter the length5
enter the width6
Area of  the rectangle: 30
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 7.dart
enter the  radiu.8
Area of the circle: 200.96
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

## Question:8

```dart
8.dart > main
Run | Debug
void main(){
stdout.write("enter the amount in shekels");
int shekels =int.parse(stdin.readLineSync()!);


  var amount_shekels=shekels*3.25;

print("Amount in dollars: ${amount_shekels} ");
}
```

```
Problems  Output  Debug Console  Terminal  Ports        powershell

 Result: 5*5 = 25
 Result: 5/5 = 1.0
 Result: 5%5 = 0
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 6.dart
enter the length5
enter the width6
Area of  the rectangle: 30
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 7.dart
enter the  radiu.8
Area of the circle: 200.96
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 8.dart
enter the amount in shekels30
Amount in dollars: 97.5
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

## Question:9

9.dart > main

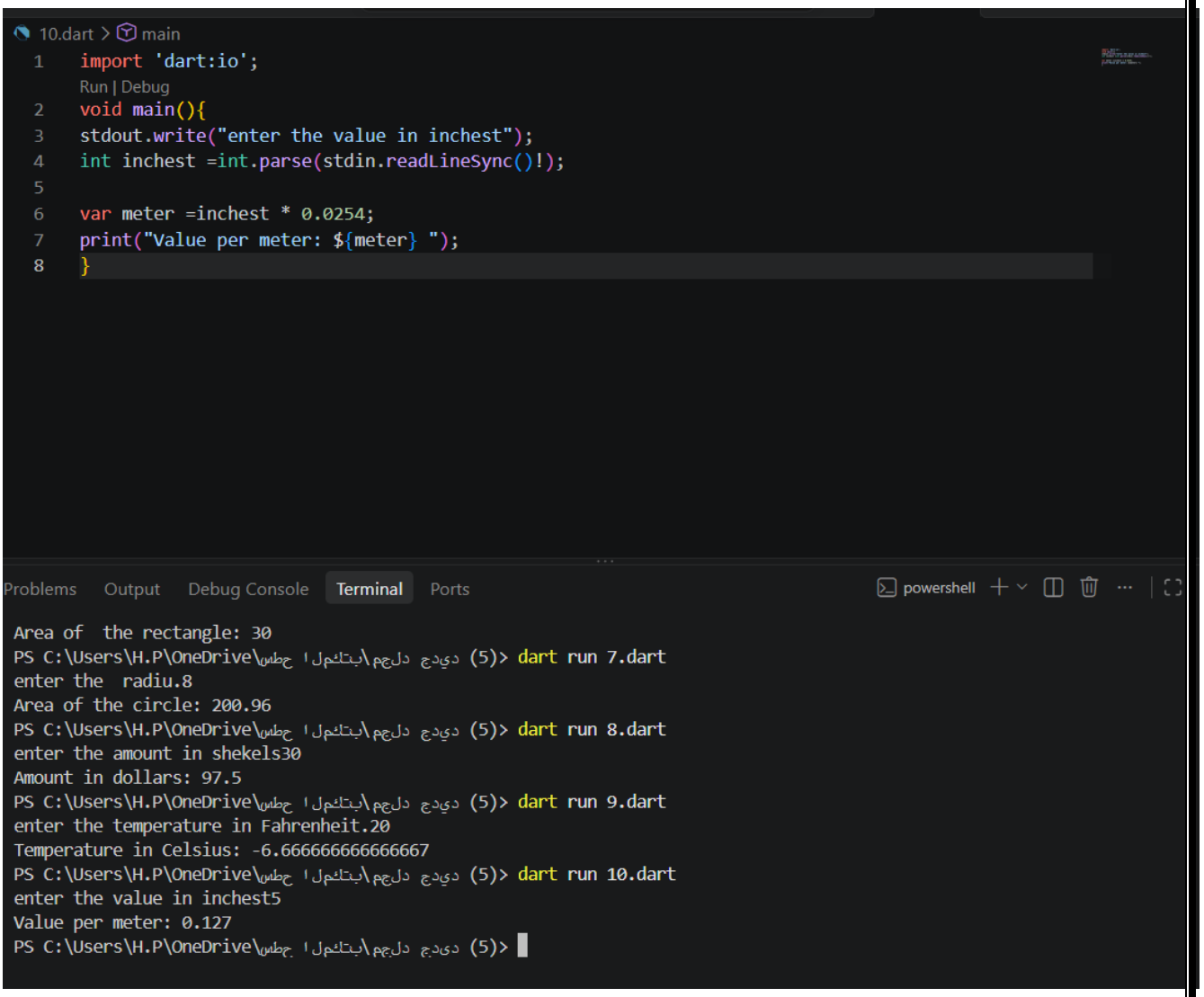
```dart
import 'dart:io';
Run | Debug
void main(){
stdout.write("enter the temperature in Fahrenheit.");
int Fahrenheit =int.parse(stdin.readLineSync()!);

var Celsius =5*(Fahrenheit-32) / 9;
print("Temperature in Celsius: ${Celsius} ");
}
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 6.dart
enter the length5
enter the width6
Area of  the rectangle: 30
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 7.dart
enter the  radiu.8
Area of the circle: 200.96
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 8.dart
enter the amount in shekels30
Amount in dollars: 97.5
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 9.dart
enter the temperature in Fahrenheit.20
Temperature in Celsius: -6.666666666666667
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

## Question:10


10.dart > main
1 import 'dart:io';
Run | Debug
2 void main(){
3 stdout.write("enter the value in inchest");
4 int inchest =int.parse(stdin.readLineSync()!);
5
6 var meter =inchest * 0.0254;
7 print("Value per meter: ${meter} ");
8 }
Problems Output Debug Console Terminal Ports
powershell
Area of the rectangle: 30
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 7.dart
enter the radiu.8
Area of the circle: 200.96
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 8.dart
enter the amount in shekels30
Amount in dollars: 97.5
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 9.dart
enter the temperature in Fahrenheit.20
Temperature in Celsius: -6.666666666666667
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 10.dart
enter the value in inchest5
Value per meter: 0.127
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>

## Question:11

11.dart > main

```dart
import 'dart:io';

Run | Debug
void main() {
  stdout.write('Enter an integer between 0 and 1000: ');
  int number = int.parse(stdin.readLineSync()!);

  int units = number % 10;
  var tens = (number / 10) % 10;
  var hundreds = (number / 100) % 10;
  var thousands = number / 1000;


  var sum = units + tens + hundreds + thousands;

  print('The sum of digits is: $sum');
}
```

Problems Output Debug Console Terminal Ports powershell

```
Area of the circle: 200.96
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 8.dart
enter the amount in shekels30
Amount in dollars: 97.5
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 9.dart
enter the temperature in Fahrenheit.20
Temperature in Celsius: -6.666666666666667
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 10.dart
enter the value in inchest5
Value per meter: 0.127
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 11.dart
Enter an integer between 0 and 1000: 100
The sum of digits is: 1.1
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

## Question:12

12.dart > main

```dart
import 'dart:io';

Run | Debug
void main(){
stdout.write("enter the number of minutes");
int minutes=int.parse(stdin.readLineSync()!);

var days= minutes/1440;
var years=minutes/525600;
print("Number of days: ${days}");
print("Number of years: ${years}");
}
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 9.dart
enter the temperature in Fahrenheit.20
Temperature in Celsius: -6.666666666666667
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 10.dart
enter the value in inchest5
Value per meter: 0.127
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 11.dart
Enter an integer between 0 and 1000: 100
The sum of digits is: 1.1
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 12.dart
enter the number of minutes5000
Number of days: 3.4722222222222223
Number of years: 0.009512937595129375
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

## Question:13

13.dart > main

```dart
import 'dart:io';

Run | Debug
void main() {
  stdout.write("enter the first nambre");
  int number1 = int.parse(stdin.readLineSync()!);

  stdout.write("enter the scond numbre");

  int number2 = int.parse(stdin.readLineSync()!);
  print("Enter one of the following operations");

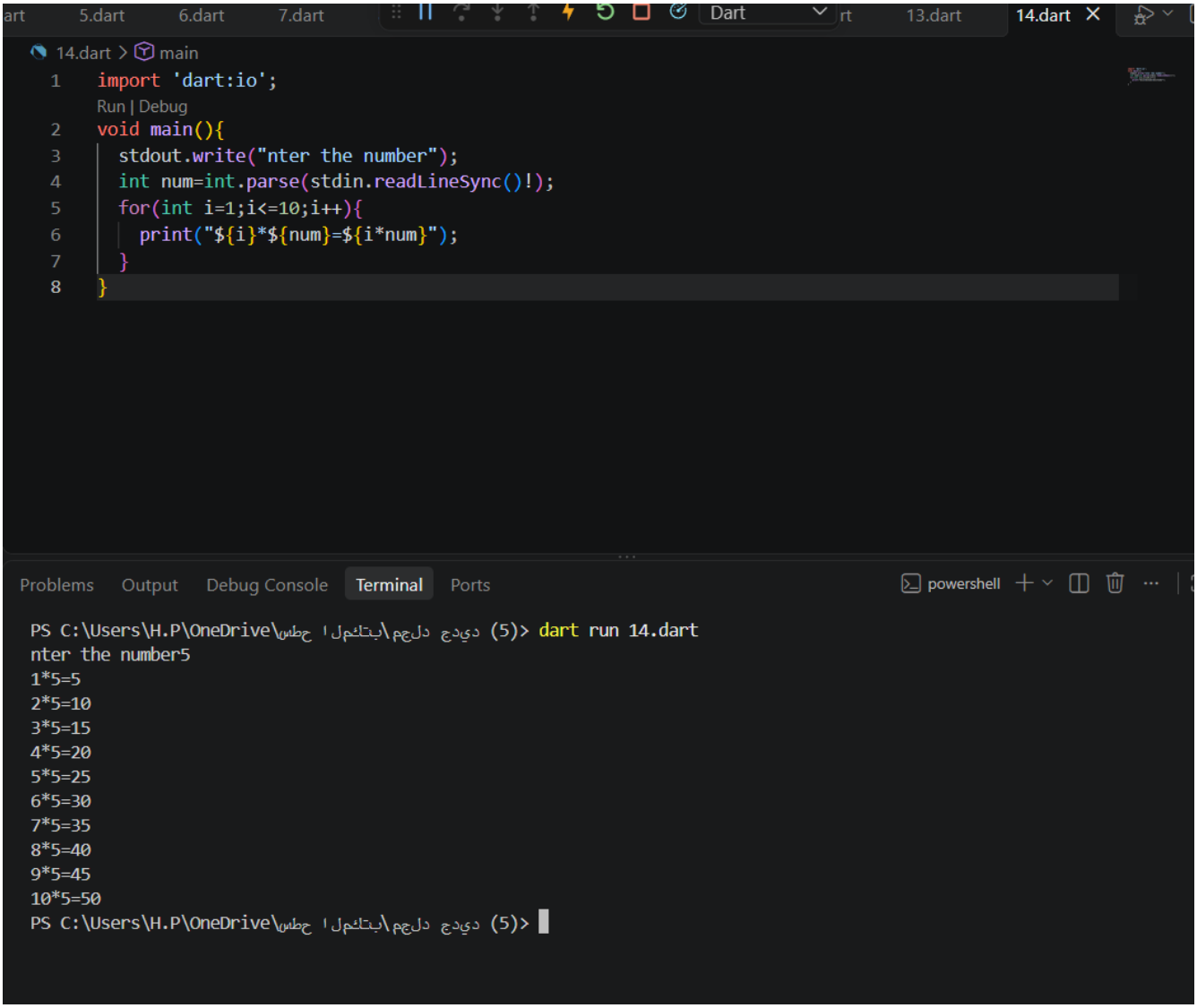
  print("+\n" + "-\n" + "*\n" + "/\n" + "%");
  stdout.write("enter the first nambre");
  String operations = (stdin.readLineSync()!);

  switch (operations) {
    case '+':
      print("${number1 + number2}");
      break;
    case '-':
      print("${number1 - number2}");
      break;
    case '*':
```

Problems Output Debug Console Terminal Ports powershell

```
enter the scond numbre6
Enter one of the following operations
+
-
*
/
%
enter the first nambre+
11
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

## Question:14


art 5.dart 6.dart 7.dart Dart rt 13.dart 14.dart
14.dart > main
1 import 'dart:io';
Run | Debug
2 void main(){
3 stdout.write("nter the number");
4 int num=int.parse(stdin.readLineSync()!);
5 for(int i=1;i<=10;i++){
6 print("${i}*${num}=${i*num}");
7 }
8 }
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 14.dart
nter the number5
1*5=5
2*5=10
3*5=15
4*5=20
5*5=25
6*5=30
7*5=35
8*5=40
9*5=45
10*5=50
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>

## Question:15

15.dart > main

```dart
import 'dart:io';

Run | Debug
void main() {
  int a = 0;
  int sum = 0;
  int x=0;
  while (a < 10) {
    stdout.write("enter the number ${a+1} : ");
    int num = int.parse(stdin.readLineSync()!);
    if (num > 0) {
      sum += num;
      a++;
    } else {
      print("Error: The number is zero or negative");
      x++;
    }
  }
  print("Final Total: ${sum}");
  print("Number of incorrect entry attempts:${x}");
}
```

Problems Output Debug Console Terminal Ports powershell

```
enter the number 4 : 5
enter the number 5 : 5
enter the number 6 : 5
enter the number 7 : 9
enter the number 8 : 9
enter the number 9 : -4
Error: The number is zero or negative
enter the number 9 : 8
enter the number 10 : 8
Final Total: 63
Number of incorrect entry attempts:2
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

# Question:16

16.dart > main

```dart
import 'dart:io';
Run | Debug
void main(){
  String userName = "alaa@gmali.com";
  String password= "123456789";
  int a=0;

  do{
    stdout.write("enter the user Name");
  String useruame = (stdin.readLineSync()!);

    stdout.write("enter thepassword");
  String passwor = (stdin.readLineSync()!);

  if(userName==useruame && password==passwor){
    print("Logged in successfully");
    break;

  }else{
    print("Try again");
    a++;
  }
  }while(a<=3);
```

Problems Output Debug Console Terminal Ports powershell

```
enter the first user Namealaa@gmali.com
enter the first password123456789
Logged in successfully
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 16.dart
enter the user Namef
enter thepassword12
Try again
enter the user Namealaa@gmali.com
enter thepassword123456789
Logged in successfully
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

## Question:17

17.dart > main

```dart
void main() {
  int grades = 5;
  double sum = 0;
   int attempts = 3;
    int gra = 0;

  for (int i = 1; i <= grades; i++) {

    while (attempts > 0) {
      stdout.write("Enter mark $i: ");
      gra = int.parse(stdin.readLineSync()!);


      if (gra >= 0 && gra <= 100) {
        break;
      } else {
        attempts--;
        print("Invalid input! You have $attempts attempts left.");
      }
    }

    if (attempts == 0 ) {
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 18.dart
enter the temperature5
cold, wear a jacket
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 17.dart
Enter mark 1: 80
Enter mark 2: 86
Enter mark 3: 90
Enter mark 4: 77
Enter mark 5: 8
The student's average is: 68.2
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

## Question:18

18.dart > main

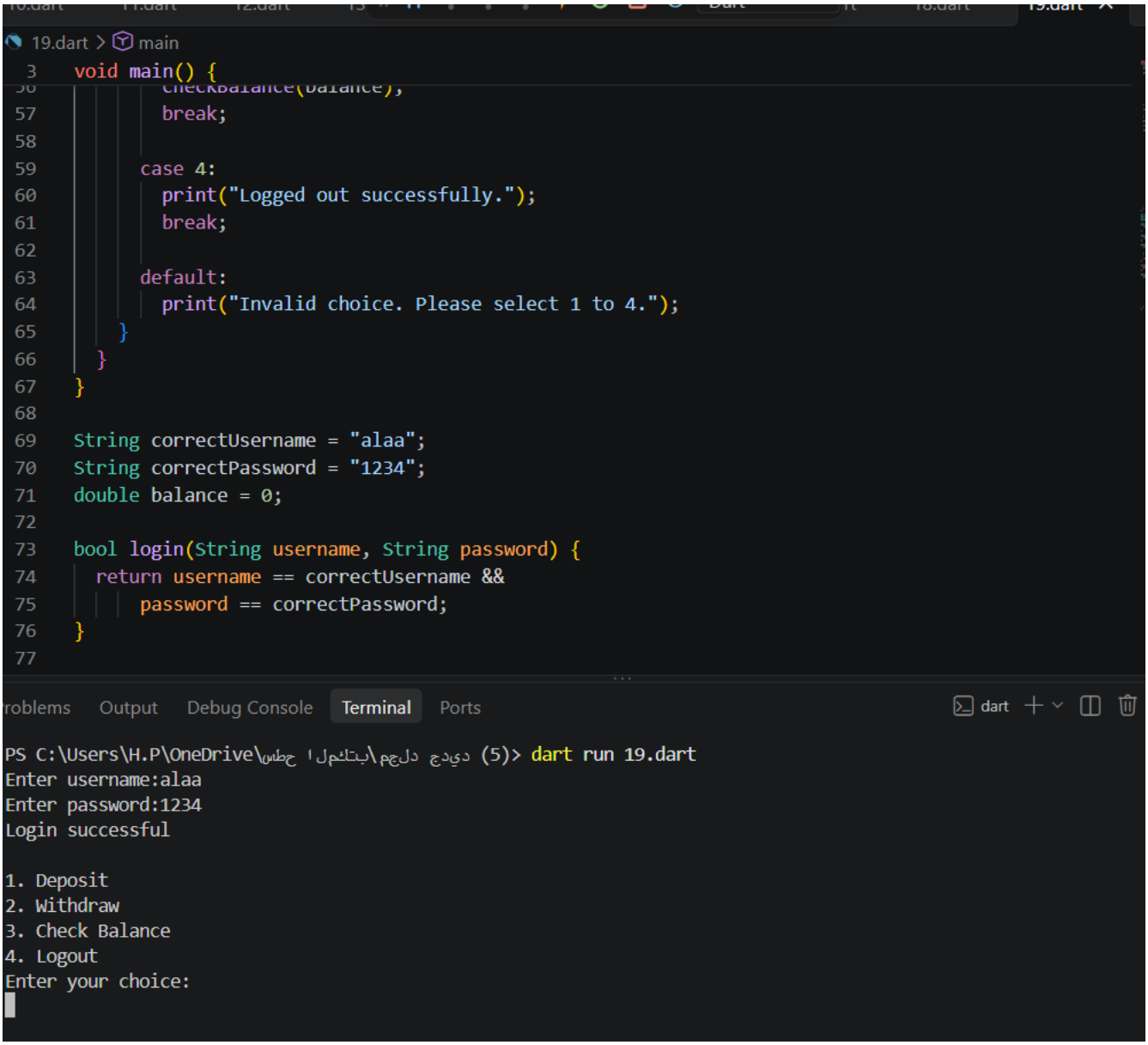
```dart
import 'dart:io';
Run | Debug
void main(){
stdout.write("enter the temperature");
int temperature=int.parse(stdin.readLineSync()!);

if(temperature<0){
  print("It's freezing, stay home");
}else if(temperature>=0&&temperature<=20){
  print("cold, wear a jacket");
}else if(temperature>=21 && temperature<=30){
  print("The weather is nice today");
}else {
  print("It's hot, stay hydrated");
}
}
```

Problems Output Debug Console Terminal Ports powershell

```
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 17.dart
Enter mark 1: 80
Enter mark 2: 86
Enter mark 3: 90
Enter mark 4: 77
Enter mark 5: 8
The student's average is: 68.2
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 18.dart
enter the temperature30
The weather is nice today
PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)>
```

## Question:19



```
19.dart > main
  3  void main() {
57            break;
58
59          case 4:
60            print("Logged out successfully.");
61            break;
62
63          default:
64            print("Invalid choice. Please select 1 to 4.");
65        }
66      }
67    }
68
69    String correctUsername = "alaa";
70    String correctPassword = "1234";
71    double balance = 0;
72
73    bool login(String username, String password) {
74      return username == correctUsername &&
75          password == correctPassword;
76    }
77
roblems  Output  Debug Console  Terminal  Ports                    dart

PS C:\Users\H.P\OneDrive\سطح المكتب\مجلد جديد (5)> dart run 19.dart
Enter username:alaa
Enter password:1234
Login successful

1. Deposit
2. Withdraw
3. Check Balance
4. Logout
Enter your choice:
```